

SLOKA THE SCHOOL

GRADE X — MATHEMATICS
WORKSHEET-1 (2026-27)
REAL NUMBERS

SECTION - A

I. Choose the correct answer type questions:

1. If a and b are two consecutive natural numbers, then the HCF (a , b) is

- A. 1
- B. 2
- C. ab
- D. $a + b$

2. If two positive integers a and b are written as $a = x^3y^2$ and $b = xy^3$; x , y are prime numbers, then HCF (a , b) is

- A. xy
- B. xy^2
- C. x^3y^3
- D. x^2y^2

3. In a Formula racing competition, the time taken by two racing cars A and B to complete one round of the track is 30 minutes and p minutes respectively. If the cars meet again at the starting point for the first time after 90 minutes and the $\text{HCF}(30, p) = 15$, then the value of p is

- A. 45 minutes
- B. 60 minutes
- C. 75 minutes
- D. 180 minutes

4. If HCF of 65 and 117 is expressible in the form $65m - 117$, then the value of m is

- A. 4
- B. 2
- C. 8
- D. 6

5. What is the largest number that divides 245 and 1029, leaving remainder 5 in each case?

- A. 16
- B. 15
- C. 9
- D. 5

6. The LCM of two co-prime numbers is always the

- A. Sum of the numbers
- B. Difference of the numbers
- C. Product of the numbers
- D. 1

7. On MG road, three consecutive traffic lights change after 36, 42 and 72 seconds. If the lights are first switched on at 9:00 am, at what time will they change simultaneously?
- A. 9:08:04
 - B. 9:08:24
 - C. 9:08:44
 - D. None of these
8. What is the smallest number which when increased by 6 becomes divisible by 36, 63 and 108?
- A. 750
 - B. 752
 - C. 754
 - D. 756
9. If N is the sum of first 13986 prime numbers, then N is always divisible by
- A. 6
 - B. 4
 - C. 8
 - D. None of these
10. If $(-1)^n + (-1)^{4n} = 0$, then n is
- A. Any positive integer
 - B. Any negative integer
 - C. Any odd natural number
 - D. Any even natural number
11. When 2^{256} is divided by 17, the remainder would be
- A. 1
 - B. 14
 - C. 16
 - D. None of these
12. If n is any natural number, then $9^{2n} - 4^{2n}$ is always divisible by
- A. 11
 - B. 4
 - C. 5
 - D. 9
13. Two numbers are in the ratio 15 : 11. If their HCF is 13, then the numbers will be
- A. 195 and 143
 - B. 190 and 140
 - C. 185 and 163
 - D. 185 and 143

14. If $A = 2n + 13$, $B = n + 7$, where n is a natural number, then HCF(A and B) is
- 2
 - 1
 - 3
 - 4
15. There are 576 boys and 448 girls in a school that are to be divided into equal sections of either boys or girls alone. The total number of sections thus formed are
- 22
 - 16
 - 36
 - 21
16. The HCF and LCM of two numbers are 33 and 264 respectively. When the first number is completely divided by 2, the quotient is 33. The other number is
- 66
 - 132
 - 232
 - 332
17. The largest number which divides 615 and 963 leaving remainder 6 in each case
- 78
 - 87
 - 67
 - 76
18. Assertion (A): If LCM of two numbers is 2475 and their product is 12375, then their HCF is 5.
Reason (R): $\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$.
- Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.
19. Assertion (A): $\sqrt{7}$ is an irrational number.
Reason (R): A square root of a prime number is always an irrational number.
- Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.
20. Assertion (A): Unit digit of 3^n cannot be an even number for any natural number n .
Reason (R): 2 is a prime factor of 3^n for any number n .
- Both (A) and (R) are true and (R) is the correct explanation of (A).
 - Both (A) and (R) are true but (R) is not the correct explanation of (A).
 - (A) is true but (R) is false.
 - (A) is false but (R) is true.

SECTION - B

II. Short answer type questions:

21. LCM of two numbers is 10 times of their HCF. Sum of HCF and LCM is 495. If one number is 90, then find the other number.
22. A forester wants to plant 66 apple trees, 88 banana trees and 110 mango trees in equal rows (in terms of number of trees). Also he wants to make distinct rows of trees (i.e., only one type of trees in one row). Find the number of minimum rows required.
23. Prove that $\sqrt{2}$ is an irrational number.
24. Prove that $2 + 5\sqrt{3}$ is an irrational number, given that $\sqrt{3}$ is an irrational number.
25. Write whether $(2\sqrt{45} + 3\sqrt{20})/(2\sqrt{5})$, on simplification, gives an irrational number or rational number.
26. Find a rational number between $\sqrt{2}$ and $\sqrt{3}$.
27. Three numbers are in the ratio 2 : 5 : 7. Their LCM is 490. Find the square root of the largest number.
28. The greatest number that will divide 76, 112, 172, and 184 so as to leave remainder 40 in each case is $k^2 \times 3$. Find the value of k.
29. Find the LCM and HCF of 2.5, 0.5 and 0.175.
30. Can we have any $n \in \mathbb{N}$, where 4^n ends with the digit zero?
31. LCM of two numbers is 14 times of their HCF. Sum of HCF and LCM is 600. If one number is 280, then find the other number.
32. Find the largest number that divides 2053 and 967 and leaves a remainder of 5 and 7 respectively.
33. Can two numbers have 18 as their HCF and 380 as their LCM? Give reason.
34. Show that 21^n cannot end with the digit 0, 2, 4, 6 and 8 for any natural number n.
35. An army contingent of 1000 members is to march behind an army band of 56 members in a parade. The two groups are to march in the same number of columns. What is the maximum number of columns in which they can march?

SECTION - C

III. Long Answer type Questions:

36. Prove that $\sqrt{3} + \sqrt{5}$ is an irrational number.
37. What is the least multiple of 7 which, when divided by 4, 12 and 16, leaves remainder 3 in each case?
38. Prove that $2\sqrt{6} - 3\sqrt{5}$ is an irrational number.
39. Prove that $\sqrt{p} + \sqrt{q}$ is an irrational number.
40. Let a and b be two positive integers. Show that $\sqrt{2}$ always lies between a/b and $(a + 2b)/(a + b)$.

SECTION - D

IV. Case Base Type Questions:

41. To enhance the reading skills of grade X students, the school nominates you and two of your friends to set up a class library. There are two sections—section A and section B of grade X. There are 32 students in section A and 36 students in section B.

(i) What is the minimum number of books you will acquire for the class library, so that they can be distributed equally among students of Section A or Section B?

(ii) If the product of two positive integers is equal to the product of their HCF and LCM, then find the $\text{HCF}(32, 36)$.

(iii) If p and q are positive integers such that $p = a$ and $q = b$, where a, b are prime numbers, then find the $\text{LCM}(p, q)$.

42. Ravish runs a book shop at school of Math, Gurgaon. He received 480 chemistry books, 192 physics books and 672 mathematics books of class XI. He wishes to arrange these books in minimum number of stacks such that each stack consists of the books of only one subject and the number of books in each stack is the same.

- a. Find the number of books in each stack.
- b. Find the number of stacks of Mathematics books.
- c. Find the minimum number of stacks of all the books.
- d. Find the difference in number of stacks of Mathematics books and sum of stacks of Physics and Chemistry books.